SURVEYING & LEVELLING · NOTE SL09

Areas and Volumes — Triangles, Trapezoidal Rule, Simpson's Rule and Earthwork

Course topics: SWCE1.4 L13

Areas of land

Divide the field into triangles and trapeziums and add their areas.

$$\text{Triangle: } A = \tfrac{1}{2} \times \text{base} \times \text{height}$$

$$\text{Trapezium: } A = \tfrac{1}{2} \text{ (sum of parallel sides)} \times \text{distance between the parallel sides}$$

Curved boundaries — offsets from a base line

Take offsets $h_0, h_1, h_2, \ldots h_n$ to the curved boundary at a common spacing d along the base line.

Trapezoidal rule

$$A = d\,(h_0/2 + h_1 + h_2 + \ldots + h_{n-1} + h_n/2)$$

- If the boundary starts and ends on the base line ($h_0 = h_n = 0$): A = d × Σh.
- It assumes straight lines between offset ends and works for any number of offsets.

Simpson's rule

$$A = (d/3)\,[h_0 + h_n + 4\,(h_1 + h_3 + \ldots + h_{n-1}) + 2\,(h_2 + h_4 + \ldots + h_{n-2})]$$

- The total number of intervals must be **even** (an odd number of offsets). Offsets may lie on both sides of the base line.
- It assumes the boundary between three successive offsets is a parabola, so it is more accurate than the trapezoidal rule.

Which offsets get 4 and which get 2? Counted from h_0, the ×4 offsets are h_1, h_3, h_5 … (odd subscripts) and the ×2 offsets are the interior h_2, h_4 … (even subscripts). The ICAR eCourse numbers the first offset O_1, so in its words the ×4 offsets are the "even ordinates" O_2, O_4 … — the same offsets. The source notes' wording "odd offsets 1st, 3rd, 5th" is ambiguous; use the subscripts above.

Offsets at 10 m spacing: 0, 2.5, 3.5, 5.0, 4.6, 3.2, 0 m (6 intervals).
Trapezoidal: A = 10 × (2.5 + 3.5 + 5.0 + 4.6 + 3.2) = 10 × 18.8 = 188 m².
Simpson: A = (10/3)[0 + 0 + 4(2.5 + 5.0 + 3.2) + 2(3.5 + 4.6)] = (10/3)(42.8 + 16.2) = 196.67 m².

Volumes of earthwork

End-area method

Use it for earth excavation and earth fill. Assume the volume between two successive cross-sections equals the average of their end areas times the distance between them:

$$V = l \times (A_1 + A_2) \,/\, 2$$

- V = volume of earthwork (m³); l = distance between cross-sections (m); A_1, A_2 = end areas of the cross-sections (m²).

Volume between two contours

$$\text{Volume} = (A_1 + A_2)/2 \times \text{contour interval}$$

A_1 and A_2 are the areas enclosed by two successive contours — the basis for computing the capacity of a pond or reservoir from a contour map.

Sources: ICAR eCourse 'Surveying and Leveling' — course content by Dr D. Tamil Mani (Dept. of Soil & Water Conservation Engg., AEC&RI, TNAU Coimbatore), reviewed by Dr V. M. Abdul Hakkim (KCAET, Kerala Agricultural University); published as an AgriMoon e-book. (Lesson 14) Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Simpson's rule needs

(a) an even number of offsets
(b) an odd number of intervals
(c) an even number of intervals
(d) equal offsets

Q2. In Simpson's rule (offsets $h_0 \ldots h_n$), the offsets multiplied by 4 are

(a) h_0 and h_n
(b) h_1, h_3, $h_5 \ldots$
(c) h_2, h_4, $h_6 \ldots$
(d) all intermediate offsets

Q3. If $h_0 = h_n = 0$, the trapezoidal rule reduces to

(a) $A = d\Sigma h$
(b) $A = d/3\ \Sigma h$
(c) $A = 2d\Sigma h$
(d) $A = \Sigma h/d$

Q4. The end-area formula for earthwork volume is

(a) $V = l(A_1 + A_2)$
(b) $V = l(A_1 + A_2)/2$
(c) $V = l(A_1 - A_2)/2$
(d) $V = (A_1 A_2)/l$

Q5. Which rule treats the boundary between offsets as straight lines?

(a) Simpson's rule
(b) Trapezoidal rule
(c) Prismoidal rule
(d) Mid-ordinate rule only

Answers

Q1 — (c) an even number of intervals — *Even intervals = odd number of offsets.*
Q2 — (b) h_1, h_3, $h_5 \ldots$
Q3 — (a) $A = d\Sigma h$
Q4 — (b) $V = l(A_1 + A_2)/2$
Q5 — (b) Trapezoidal rule